

ДЗ.11 N1 Найдите многочлен GF(2ⁿ)
 посправе ког Page - Cononovore
 $n=15, d=5$

Т.к. $d-1=4:$

$$\begin{aligned}
 g(x) &= (x+d)(x+d^2)(x+d^3)(x+d^4) = \\
 &= x^4 + x^3(d+d^2+d^3+d^4) + x^2(d^7+d^6+d^5+d^4+d^3) + \\
 &\quad + x(d^9+d^8+d^7+d^6) + d^{10} = \\
 &\quad d+d^2=d^5 \quad d^2+d^6=d^{10} \quad ; \quad d^5+d^7=d^{13} \quad d^{10}+d^{12}=d^3 \\
 &\quad d^3+d^4=d^7 \quad d^8+d^9=d^{12} \quad ; \quad d^{10}+d^7=d^6 \\
 &= x^4 + x^3(d^{13}) + x^2(d^6) + x(d^3) + d^{10}
 \end{aligned}$$

В дальнейшем Page

$$\begin{aligned}
 d^{13} &= 1+d^2+d^3 \\
 d^6 &= d^2+d^3 \\
 d^3 &= d^3 \\
 d^{10} &= 1+d+d^2
 \end{aligned}$$

$$g(x) = x^4 + x^3(1+d^2+d^3) + x^2(d^2+d^3) + x(d^3) + (d+1+d^2)$$

$$k = n - (d-1) = 15 - 4 = 11$$

$$\boxed{(15, 11); d=5}$$

N2 (15, 9)

$$d = n - k + 1 = 15 - 9 + 1 = 7$$

$$d = 2t+1 \Rightarrow t = 3$$

$$\begin{aligned}
 g(x) &= (x+d)(x+d^2)(x+d^3)(x+d^4)(x+d^5)(x+d^6) = \\
 &= x^6 + x^5(d+d^2+d^3+d^4+d^5+d^6) + \\
 &\quad + x^4(d^3+d^4+d^5+d^6+d^7+d^8+d^9+d^{10}+d^{11}) + \\
 &\quad + x^3(d^6+d^7+d^8+d^9+d^{10}+d^{11}+d^{12}+d^{13}+d^{14}+d^{15}) + \\
 &\quad + x^2(d^{18}+d^{17}+d^{16}+d^{15}+d^{14}+d^{13}+d^{12}+d^{11}+d^{10}+d^9) + \\
 &\quad + x(d^{20}+d^{19}+d^{18}+d^{17}+d^{16}+d^{15}) + d^{21} = \\
 &= x^6 + x^5(d+d^2+d^3+d^4+d^5+d^6) + x^4(d^3+d^4+d^{10}+d^{11}) + \\
 &\quad + x^3(d^6+d^7+d^9+d^{10}+d^{12}+d^{11}+d^{14}+d^9) + x^2(d^5+d^6+d^{11}+d^{10}) + \\
 &\quad + x(d^0+d^1+d^5+d^4+d^3+d^2) + d^6 =
 \end{aligned}$$

$$= x^6 + x^5 d^{10} + x^4 d^{14} + d^4 x^3 + d^6 x^2 + d^9 x + d^6$$

$$S_j = (e(x^j) + e(d^j))$$

$$t=3 \Rightarrow j=1, 2, \dots, 6$$

Равные слова: $d d^3 d^{10} d^{12} d^{13} d^3 d^9 d^5 1 d^{10} d^8 d d^{13} d^2$

$$S_1 = d^1 + d^4 + d^{12} + 1 + d^2 + d^8 + 1 + d^{12} + d^8 + d^4 + d^3 + d^{12} + d^{13} + d^{11} + d^1 = 0$$

$$S_2 = d^1 + d^5 + d^{14} + d^3 + d^6 + d^{13} + d^6 + d^4 + d^1 + d^{13} + d^{13} + d^8 + d^{10} + d^9 + 1 = 0$$

$$S_3 = 0, \quad S_4 = 0, \quad S_5 = 0, \quad S_6 = 0$$

Нет оснований в равных словах.

$$e = e_i, \quad \text{где } i = \text{---} 14, \quad e_i = 0 \quad \forall \quad i$$

т.е. нулевой вектор размерности 14.

$$e = \underbrace{(0, 0, 0, 0, 0)}_{14 \text{ нулей}}, (0, 0)$$